

JKBOSE 2026

CLASS 11TH



Relations and Functions

MATHS

LECTURE-03

BY-MEHRAJ SIR



TOPICS TO BE COVERED

1

Exercise Discussion 2.2

2

What is Function?

3

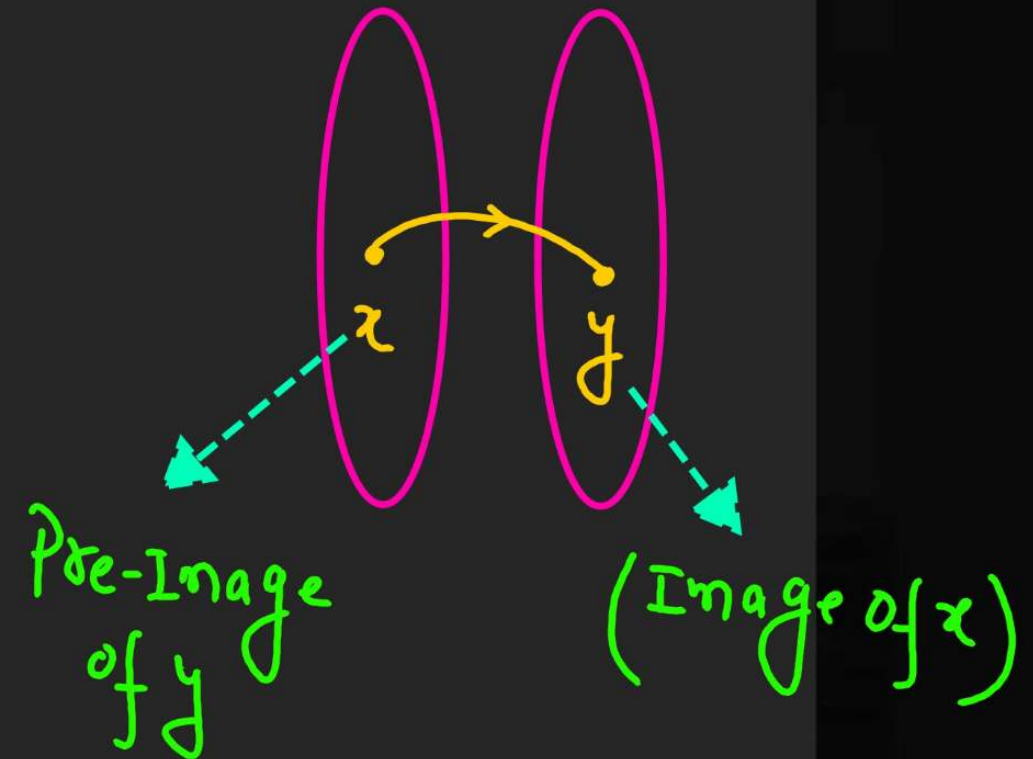
Related Problems

Relation:

$$R \subseteq A \times B$$

* Relation from A to B.

$$* 2^{A \times B}$$



EXERCISE – 2.2

Question



Let $A = \{1, 2, 3, \dots, 14\}$. Define a relation R from A to A by $R = \{(x, y) : 3x - y = 0, \text{ where } x, y \in A\}$.

Write down its domain, codomain and range.

(Sol) Given Relation $R = \{(x, y) : 3x - y = 0\}$

$$\Rightarrow R = \{(x, y) : y = 3x\}$$

$$\Rightarrow R = \{(1, 3), (2, 6), (3, 9), (4, 12)\}$$

$$\text{So Domain of } R = \{1, 2, 3, 4\}$$

$$\text{Range of } R = \{3, 6, 9, 12\}$$

$$\text{Co-Domain of } R = A.$$

Question

Define a relation R on the set N of natural numbers by $R = \{(x, y) : y = x + 5, x \text{ is a natural number less than } 4; x, y \in N\}$. Depict this relationship using roster form. Write down the domain and the range.

(Sol) Given Relation $R = \{(x, y) : y = x + 5, x < 4, x, y \in N\}$

$$\Rightarrow R = \{(1, 6), (2, 7), (3, 8)\}$$

$$\text{So, Domain of } R = \{1, 2, 3\}$$

$$\text{Range of } R = \{6, 7, 8\}$$

Question (PyQ).

$A = \{1, 2, 3, 5\}$ and $B = \{4, 6, 9\}$. Define a relation R from A to B by $R = \{x, y\}$: the difference between x and y is odd; $x \in A, y \in B\}$. Write R in roster form.

(Sol) Given Relation R is;

$$R = \{(x, y) : \text{the diff. b/w } x \text{ \& } y \text{ is odd}\}$$

$$\Rightarrow R = \{(1, 4), (1, 6), (2, 9), (3, 4), (3, 6), (5, 4), (5, 6)\}$$

Hence the result.

Question

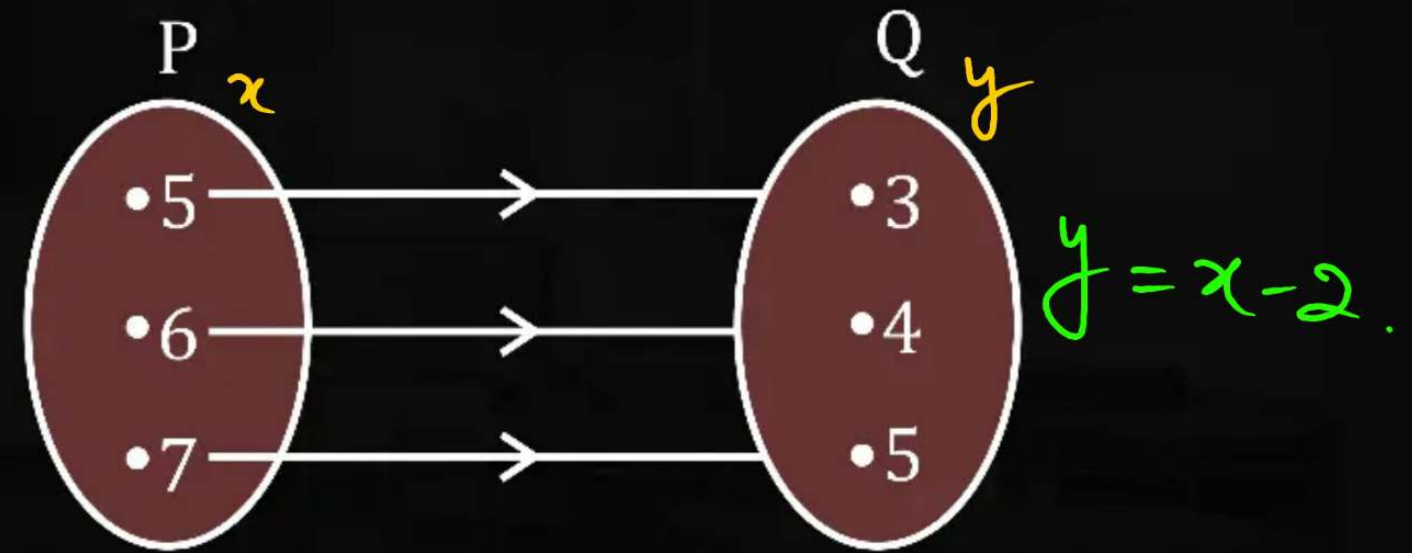
The figure shows a relationship between the sets P and Q. Write this relation

(i) In set-builder form = $\{(x, y) : y = x - 2\}$

(ii) Roster form. #

What is its domain and range?

H/W



Question

Let $A = \{1, 2, 3, 4, 6\}$. Let R be the relation on A defined by

$\{(a, b) : a, b \in A, b \text{ is exactly divisible by } a\}$.

- (i) Write R in roster form
- (ii) Find the domain of R
- (iii) Find the range of R .

(Sol) let $R = \{(a, b) : b \text{ is exactly divisible by } a\}$

$$\Rightarrow R = \left\{ (1, 1), (1, 2), (1, 3), (1, 4), (1, 6), (2, 2), (2, 4), (2, 6), (3, 3), (3, 6), (4, 4), (6, 6) \right\}$$

$$\text{So Domain} = \{1, 2, 3, 4, 6\} = A$$

$$\& \text{Range} = \{1, 2, 3, 4, 6\} = A$$

Question

Determine the domain and range of the relation R defined by

$$R = \{x, x + 5) : x \in \{0, 1, 2, 3, 4, 5\}\}.$$

H/W

Question

Write the relation $R = \{(x, x^3) : x \text{ is a prime number less than } 10\}$ in roster form.

$$R = \{(2, 8), (3, 27), (5, 125), (7, 343)\}.$$

Question

Let $A = \{x, y, z\}$ and $B = \{1, 2\}$. Find the number of relations from A to B.

$$\begin{aligned} \text{(Sol) Number of Relations} &= 2^{n(A) \times n(B)} \\ &= 2^{3 \times 2} = 2^6 = 64. \end{aligned}$$

Question

$$R = \mathbb{Z} \times \mathbb{Z}$$

Let R be the relation on $\mathbb{Z}$ defined by $R = \{(a, b), a, b \in \mathbb{Z}, a - b \text{ is an integer}\}$

Find the domain and range of R .

(Sol) Given Relation on $\mathbb{Z}$ is,

$$R = \{(a, b) : a, b \in \mathbb{Z}, a - b \in \mathbb{Z}\}$$

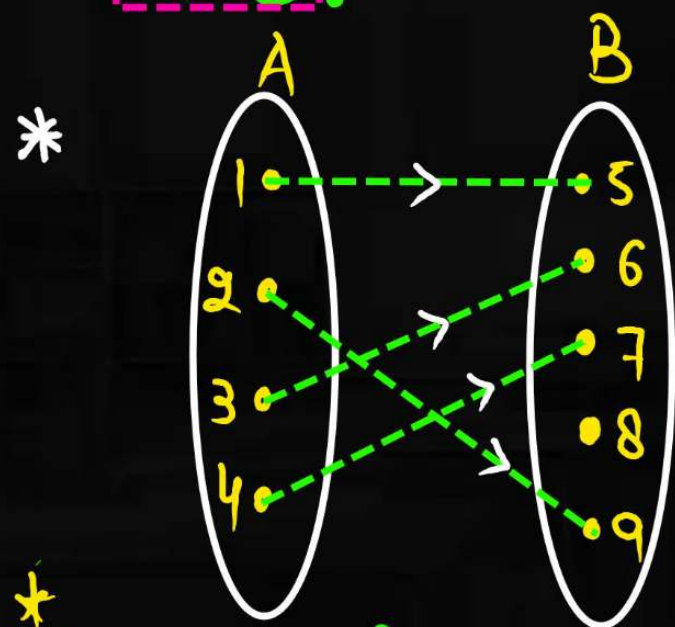
Since we know that the difference of any two integers is always an integer. So

$$\Rightarrow \text{Domain} = \mathbb{Z}, \text{Range} = \mathbb{Z}$$

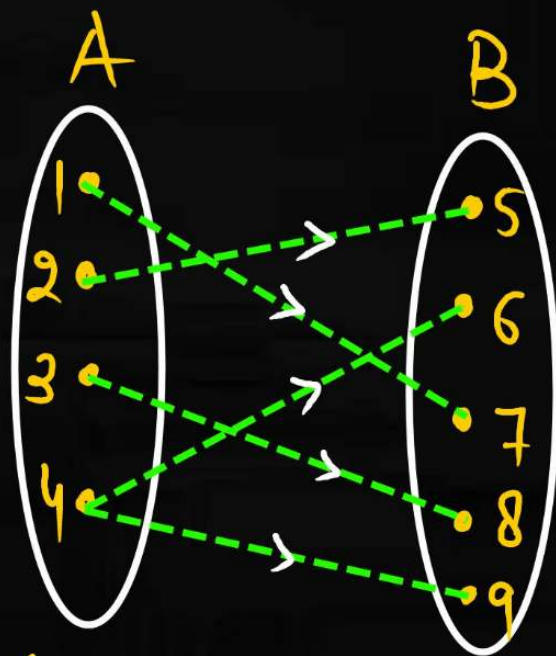
Functions

* Special Type of relation from A to B.

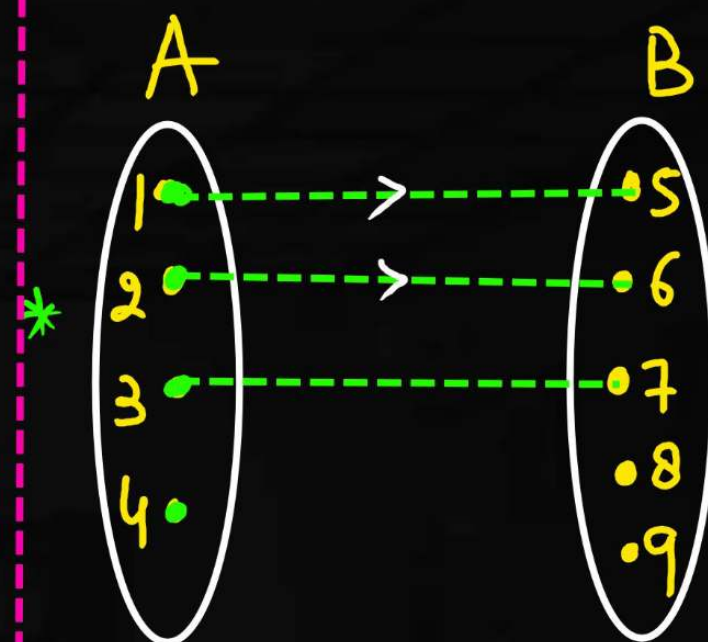
* Every element of A has a unique Image in B.



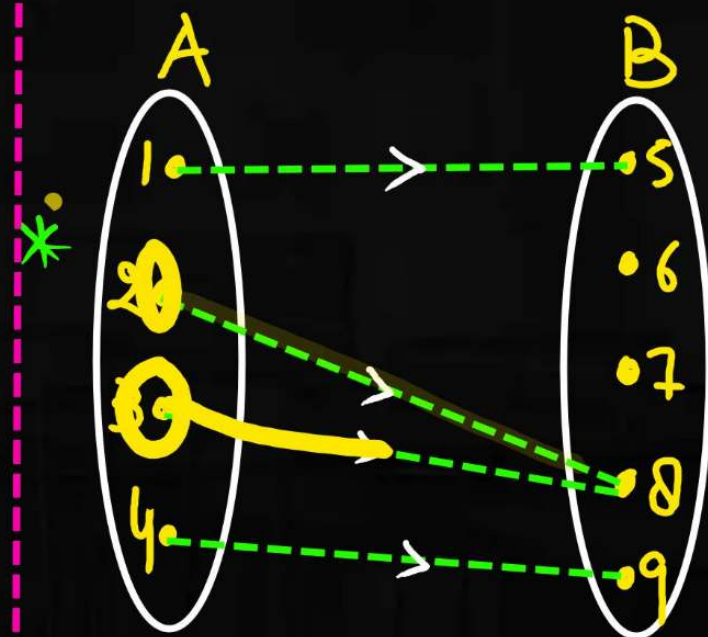
* (It is a function from A to B.)



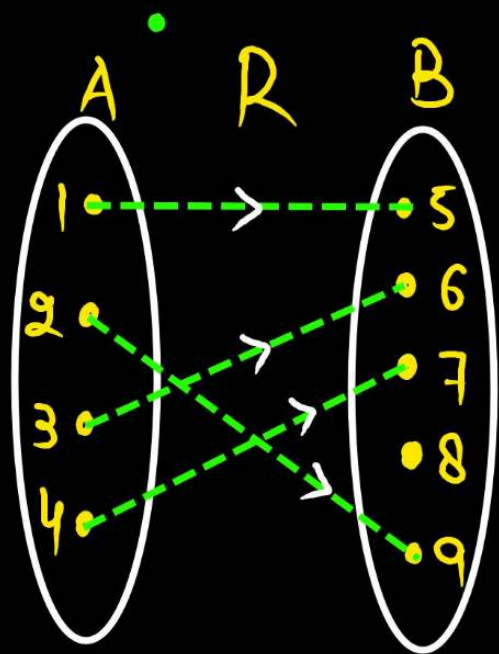
* (It is not a function from A to B)



(It is not a function from A to B)



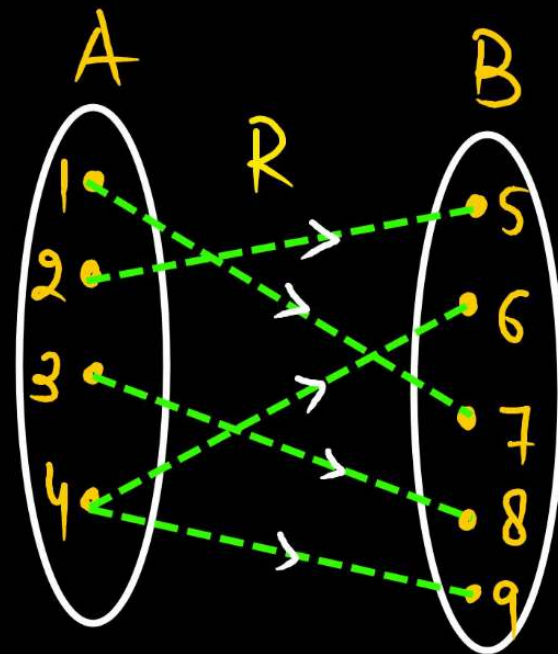
It is a function from A to B.



$$R = \{(1, 5), (2, 9), (3, 6), (4, 7)\}$$

Definition A Relation R from A to B

is said to be a function if
Domain of R is A & no two
distinct ordered pairs of R
have the same first entry.

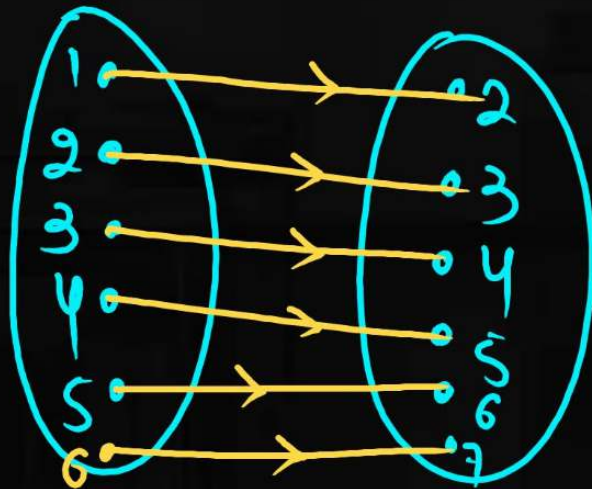


$$R = \{(1, 7), (2, 5), (3, 8), (4, 6), (4, 9)\}$$

Example

Examine each of the following relations given below and state in each case, giving reasons whether it is a function or not?

- (i) $R = \{(2, 1), (3, 1), (4, 2)\}$ (It is a function)
- (ii) $R = \{(2, 2), (2, 4), (3, 3), (4, 4)\}$ (It is not a function)
- (iii) $R = \{(1, 2), (2, 3), (3, 4), (4, 5), (5, 6), (6, 7)\}$ (It is a function)



$$* f(x) = x^2$$

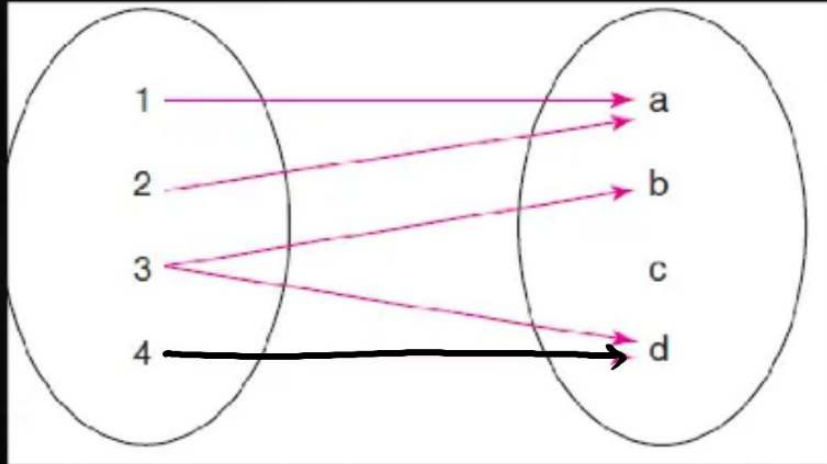
$$f(1) = 1^2 = 1$$

$$f(2) = 2^2 = 4$$

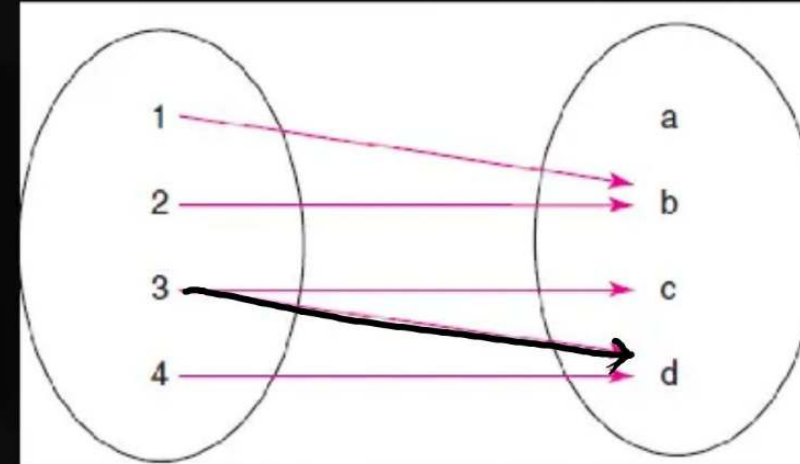
Question

Which one of the following represents a function?

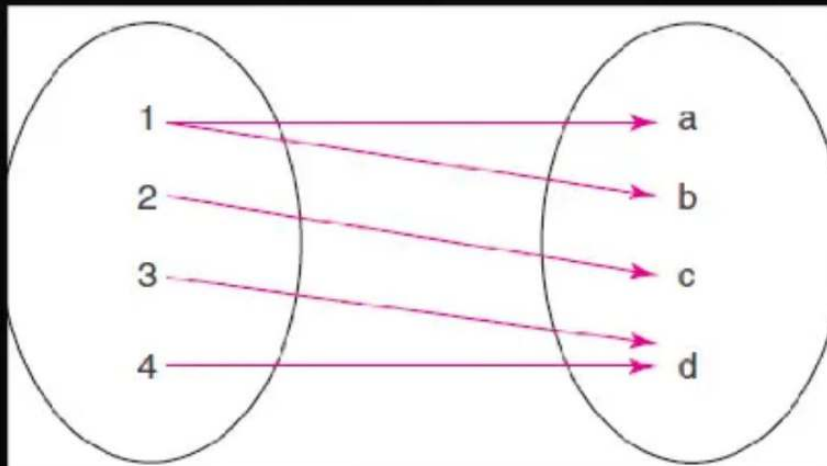
~~1~~



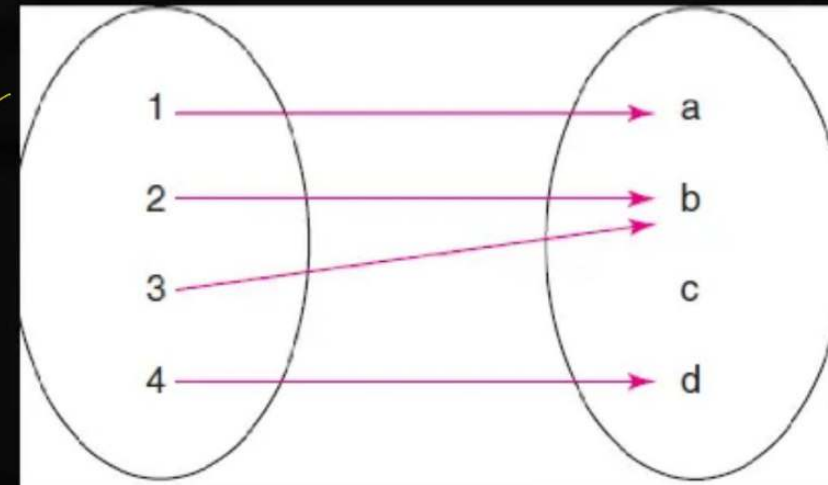
~~2~~



~~3~~



4



Question

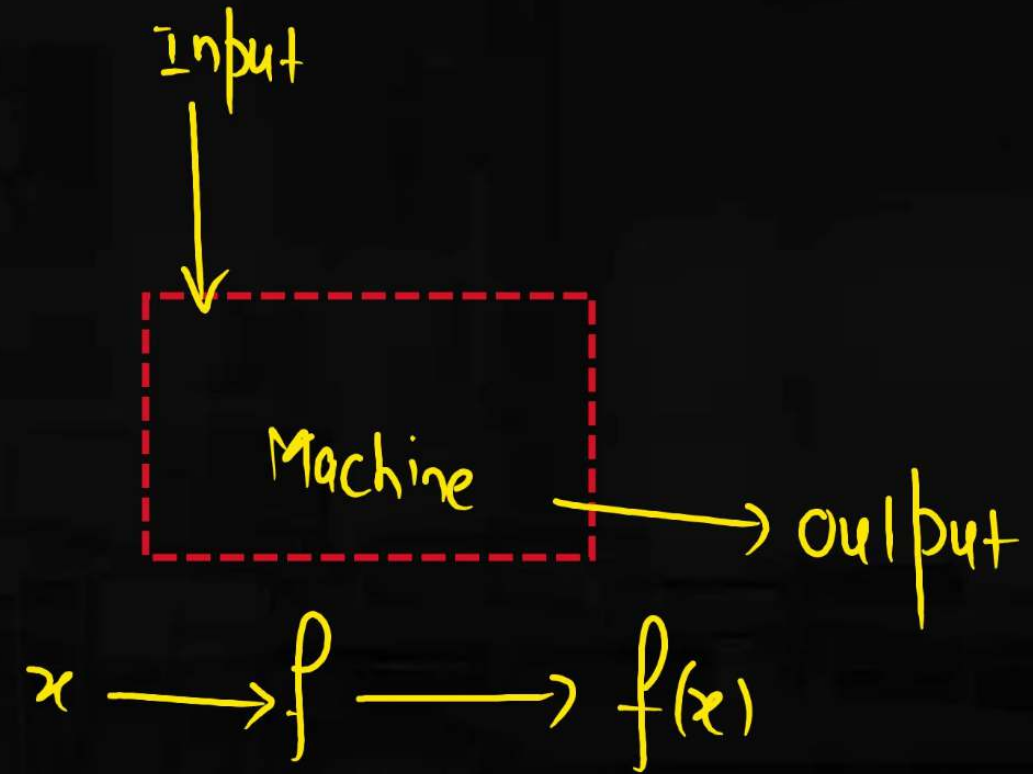
Let $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$ be two sets. Which one of the following subsets of $A \times B$ defines a function from A to B ?

1 ✓ $f_1 = \{(1, 2), (2, 3), (3, 4)\}$

2 ✗ $f_2 = \{(\underline{1}, 2), (\underline{1}, 3), (2, 3), (3, 4)\}$

3 ✗ $f_3 = \{(1, 3), (2, 4)\}$ ✗

4 ✗ $f_4 = \{(1, 4), (\underline{2}, 4), (3, 4), (\underline{2}, 3)\}$ ✗



Question

If $f(x) = x^3 - \frac{1}{x^3}$, then $f(x) + f\left(\frac{1}{x}\right)$ is equal to

☒ 1 $2x^3$

☒ 2 $2\frac{1}{x^3}$

☒ 3 0

☒ 4 1

$$f(x) = x^3 - \frac{1}{x^3} \quad \text{--- (1)}$$

$$f\left(\frac{1}{x}\right) = \frac{1}{x^3} - x^3 \quad \text{--- (2)}$$

Adding (1) & (2), we get;

$$f(x) + f\left(\frac{1}{x}\right) = \cancel{x^3} - \cancel{\frac{1}{x^3}} + \cancel{\frac{1}{x^3}} - \cancel{x^3}$$

$$\Rightarrow f(x) + f\left(\frac{1}{x}\right) = 0$$

Summary

- Function is a Special Type of relation from A to B in which every element of A has a Unique image in B
- Set of Values of x for which $f(x)$ is defined is known as the **Domain of the function.**
- Set of the values assumed by the function f is known as the **Range of the Function.**

Thank
f **You**